

Welcome to the iddoverse: An R package for converting IDDO-SDTM data into analysis datasets

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: [↗](#)

Submitted: 07 April 2026

Published: unpublished

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Summary

Study Data Tabulation Model (SDTM), developed by the Clinical Data Interchange Standards Consortium (CDISC), is an internationally adopted data storage standard ([Clinical Data Interchange Standards Consortium, 2025](#)). The model provides a coherent framework for harmonising data from multiple clinical studies. Using a modified implementation of SDTM, the Infectious Diseases Data Observatory (IDDO) has developed a standardised repository of individual participant data from clinical studies spanning diseases of global health importance ([Infectious Diseases Data Observatory, 2025](#)). Generating datasets amenable to analysis ('analysis datasets') from SDTM format requires complex transformations, which requires additional time and resource by end-users.

The iddoverse package provides a suite of functions that transforms datasets from SDTM into analysis datasets. Assuming a foundational familiarity with the R software ([R Core Team, 2022](#)), the package is intended for epidemiologists, statisticians, and data scientists with a diverse range of programming ability. Advanced knowledge of SDTM is not required, thereby removing a potential challenge for data requesters, especially those in Low- and Middle- Income Countries (LMICs) where SDTM training and expertise is often more difficult to access. Overall, the iddoverse package aims to reduce time spent on data manipulation and facilitate the generation of reproducible analysis datasets ([Patel SK, 2023](#)).

Statement of Need

In the context of infectious diseases, IDDO maximises the scientific utility of existing clinical data through the promotion of data reuse and the adoption of a responsible open-access data model ([Pisani E, 2016, 2017](#)). By pooling multiple datasets, scientists can address important questions using individual participant data meta-analysis (IPD-MA) techniques. To achieve this, IDDO standardises clinical study data from disparate sources into SDTM format, thereby providing a consistent and comprehensive method to harmonise data. The result is a controlled, open-access database, which enables the global research community to address questions of public health relevance that would otherwise not be possible using a standalone study. A key challenge, however, is the effort and time required for transformation of the curated data to an analysis-ready format. This mandates a working knowledge of the SDTM format and nomenclature, in addition to a good knowledge of IDDO's specific implementation of SDTM (a necessary modification to existing implementation guidelines due to heterogeneity in legacy data ([Infectious Diseases Data Observatory, 2019](#))). End-users unfamiliar with IDDO's SDTM implementation can therefore encounter challenges in utilising the data.

State of the Field

The challenges associated with transforming SDTM formatted data into analysis datasets are not unique to IDDO and several pre-existing tools fully, or partially, support the transformation process (Chen HL, 2010; pharmaverse, 2023; Suchchenko L, 2023). These tools were designed to work on the standard implementation of SDTM and assume a working knowledge of CDISC standards. Other tools to generate analysis datasets are either private or follow commercial business models, which are not feasible in academic and resource limited settings (ICON, n.d.; International Drug Development Institute, n.d.; Quanticate, n.d.).

The iddoverse package is tailored to IDDO's SDTM implementation and is designed to be used by those unfamiliar with the format. Additionally, the package has been developed for researchers across the globe, specifically those in LMICs, and is not developed for commercial purpose nor for pharmaceutical regulatory submission.

SDTM Data

Data stored in SDTM format comprises a series of subsets called domains, with each domain corresponding to a specific data topic. Domains are tabular and usually stored in a long format; typically a row per event per participant, which can result in multiple rows per participant, per timepoint. The package provides several synthetic datasets generated for user familiarisation and documentation. SDTM findings domains, such as the microbiology (MB) domain, can have up to 4 columns for capturing the test results (i.e. MBORRES, MBSTRESC & MBSTRESN) and a selection of over 20 timing variables. The MB domain, for example (see MB_RPTESTB below), has one row for every microbiology test and test result conducted in the study. Whilst this preserves the intricacies of the study data, it also creates complexity for analysis. The iddoverse package aims to minimise this analytical burden whilst maximising the retention of data granularity.

```
> library(iddoverse)
> library(tidyverse)
> MB_RPTESTB %>%
  select(STUDYID, USUBJID, MBTESTCD, MBORRES, MBORRESU, MBSTRESN, MBSTRESU,
         MBDY, VISITDY, everything())
# A tibble: 14 × 39
  STUDYID USUBJID MBTESTCD MBORRES MBORRESU MBSTRESN MBSTRESU MBDY VISITDY
<chr> <chr> <chr> <dbl> <chr> <dbl> <chr> <dbl> <dbl>
1 RPTESTB RPTESTB... PFALCIPA 899 10^6/L 899 10^6/L 1 1
2 RPTESTB RPTESTB... PFALCIPA 0 10^6/L 0 10^6/L 3 3
3 RPTESTB RPTESTB... PFALCIPA 450 10^6/L 450 10^6/L 85 85
4 RPTESTB RPTESTB... PFALCIPA 2987 10^6/L 2987 10^6/L 181 182
5 RPTESTB RPTESTB... PFALCIPA 66 /500 WBC 1056 10^6/L 1 1
6 RPTESTB RPTESTB... PFALCIPA 3 /500 WBC 48 10^6/L 4 3
7 RPTESTB RPTESTB... PFALCIPA 0 /500 WBC 0 10^6/L 86 85
8 RPTESTB RPTESTB... PFALCIPA 0 /500 WBC 0 10^6/L 182 182
9 RPTESTB RPTESTB... PFALCIPA 0 /500 WBC 0 10^6/L 273 273
10 RPTESTB RPTESTB... PFALCIPA 0 /500 WBC 0 10^6/L 362 366
11 RPTESTB RPTESTB... PFALCIPA 989 10^6/L 989 10^6/L 2 1
12 RPTESTB RPTESTB... PFALCIPA 0 10^6/L 0 10^6/L 5 3
13 RPTESTB RPTESTB... PFALCIPA 167 10^6/L 167 10^6/L 85 85
14 RPTESTB RPTESTB... PFALCIPA 0 10^6/L 0 10^6/L 184 182
# 30 more variables: DOMAIN <chr>, MBSEQ <dbl>, MBGRPID <lgl>,
# MBREFID <lgl>, MBTEST <chr>, MBMODIFY <lgl>, MBTSTDTL <chr>,
# MBCAT <lgl>, MBSTRESC <dbl>, MBSTAT <lgl>, MBREASND <lgl>, MBNAM <lgl>,
# MBSPEC <chr>, MBSPCCND <lgl>, MBLOC <chr>, MBMETHOD <chr>,
```

```
# VISITNUM <dbl>, VISIT <chr>, EPOCH <chr>, MBDTCT <chr>, MBTPT <lgl>,  
# MBTPTNUM <lgl>, MBELTM <lgl>, MBTPTREF <lgl>, MBSTRF <lgl>,  
# MBEVINTX <lgl>, MBSTRTPT <lgl>, MBSTTPT <lgl>, MBCDSTDY <lgl>, ...
```

Software Design

The guiding design philosophy has been to promote greater confidence amongst researchers using data from the IDDO repository by semi-automating the data transformation from SDTM format into analysis datasets. As a result, a greater proportion of the scientific community can benefit from the powerful database IDDO has built over the past 15 years and contribute towards the global fight against infectious diseases.

The `iddoverse` suite (Figure 1) comprises several functions. Most functions are domain-agnostic and can be applied across special purpose, findings, and event domains. This approach contrasts with the earlier developmental versions of the package (pre version 0.7.0) which had predominately domain-specific functions. Previously, a static selection of SDTM timing variables was used, which proved to be too restrictive and impacted the generalisability of the functions, so a customisable set has now been implemented.

A key limitation is that the `iddoverse` functions cannot address every need of researchers due to the large variability in the datasets within, and across, diseases. The objective has been to provide assistance and automation of analysis datasets, whilst keeping the solution generalisable and customisable by the user.

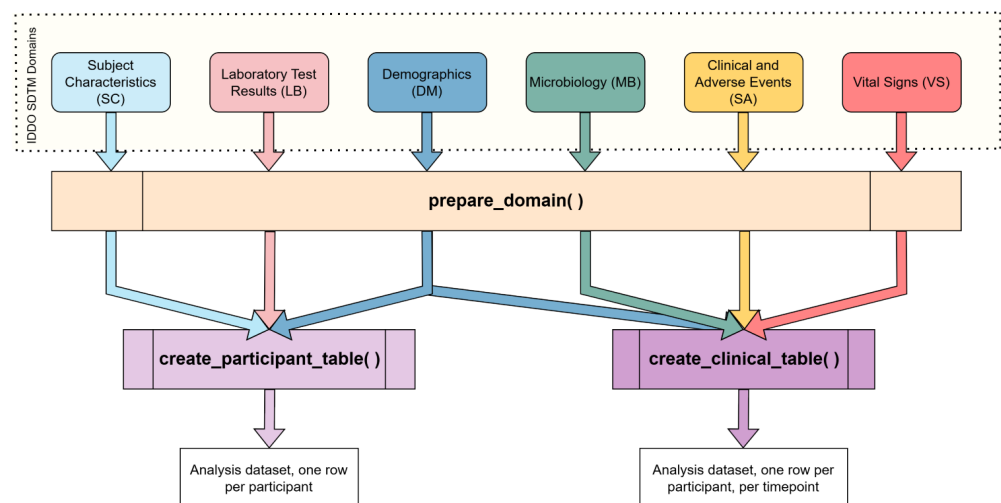


Figure 1: Figure 1: Flowchart of functions within the `iddoverse` package.

iddoverse Functions

A core function within the `iddoverse` package is `prepare_domain()`. This function enables the transformation of a single IDDO-SDTM domain. In order to reduce the number of result and timing columns, the function amalgamates data into one 'best choice' for 'time' and 'result'. For the 'best result', the standardised numeric result (i.e. `MBSTRESN`) is taken first for each row. If this standardised numeric result is missing for a given row, the standardised character value (i.e. `MBSTRESC`) will instead be populated as the best choice result for that given row (Figure 2). If both standardised results are absent from a row, the modified result (i.e. `MBMODIFY`) will be used, where present. The original contributed result from the study (i.e. `MBORRES`) is

90 utilised if no other options are available. Note that MBMODIFY is used by IDDO specifically for
91 a handful of diseases and will not be relevant to most. The hierarchy of timing variables is
92 study dependent – it is therefore a customisable parameter to enable researchers to select the
93 most appropriate variable(s) for their analysis. By choosing a ‘best choice’ timing and result,
94 potential confusion surrounding multiple columns is removed.

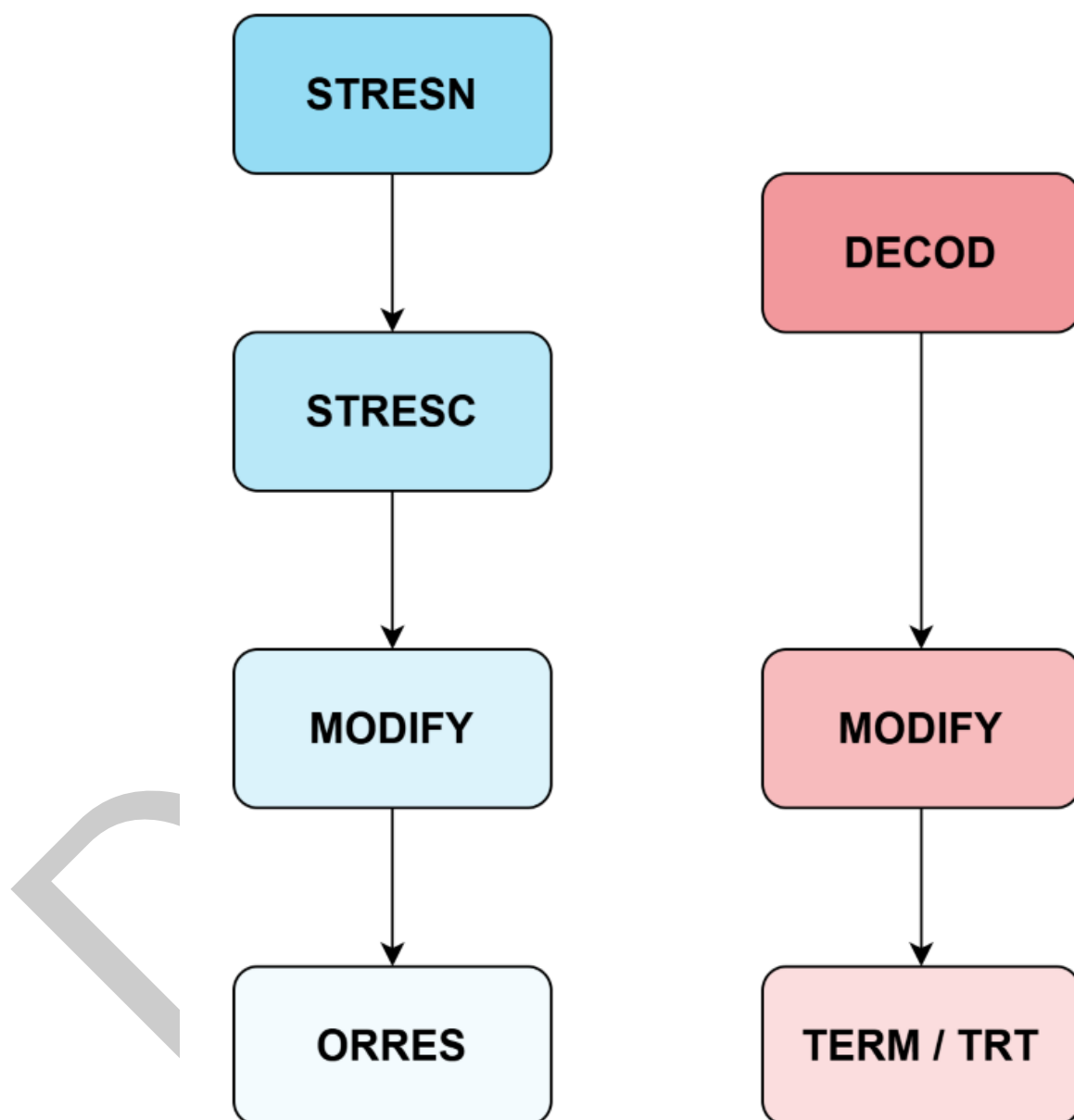


Figure 2: Figure 2: Hierarchy of best choice results/events/findings in `prepare_domain()`. STRESN or DECOD would be used in the first instance and, where rows are missing this information, they are populated with the variables under them in order. The two letter domain code precedes these variable names.

95 The `prepare_domain()` function then pivots the rows by the best choice time variable (TIME,
96 TIME_SOURCE), the study ID (STUDYID) and participant number (USUBJID). The different

97 events/findings/tests are transformed into columns, and the dataset is populated with the
 98 associated result. Several domains can then be analysed separately or joined together by the
 99 uniquely identifying keys: STUDYID, USUBJID, TIME and TIME_SOURCE.

```
> prepare_domain(MB_RPTESTB, "mb")
[1] "Number of rows where values_fn has been used to pick record in the MB domain: 0"
# A tibble: 14 × 5
  STUDYID USUBJID      TIME TIME_SOURCE `PFALCIPA_10^6/L`
  <chr>   <chr>    <chr> <chr>      <chr>
1 RPTESTB RPTESTB_001 1      DY        899
2 RPTESTB RPTESTB_001 3      DY         0
3 RPTESTB RPTESTB_001 85     DY       450
4 RPTESTB RPTESTB_001 181    DY     2987
5 RPTESTB RPTESTB_002 1      DY     1056
6 RPTESTB RPTESTB_002 4      DY        48
7 RPTESTB RPTESTB_002 86     DY         0
8 RPTESTB RPTESTB_002 182    DY         0
9 RPTESTB RPTESTB_002 273    DY         0
10 RPTESTB RPTESTB_002 362    DY         0
11 RPTESTB RPTESTB_003 2      DY     989
12 RPTESTB RPTESTB_003 5      DY         0
13 RPTESTB RPTESTB_003 85     DY     167
14 RPTESTB RPTESTB_003 184    DY         0
```

100 Parameters provide customisation options such as variable selection, inclusion of information
 101 for test methodology or location, and the mechanism for handling rows that are not uniquely
 102 separable in the pivot process.

103 Additionally, the package contains functions to create standardised, multiple domain analysis
 104 datasets, such as `create_participant_table()`. These functions merge various domain
 105 analysis datasets together by using the `prepare_domain()` function multiple times to extract
 106 specific variables from their source domains. The choice of which variables to include is
 107 based on subject matter expertise. The purpose of these tables is to provide most of the key
 108 information required for analyses, for instance the information typically presented in Table 1 of
 109 clinical study publications.

110 `iddoverse` also provides functions that check and summarise data, which confers support for
 111 exploratory data analysis and informs the selection of parameters within `prepare_domain()`.
 112 Additionally, some utility and support functions exist to derive or convert certain variables.

113 Research Impact Statement

114 The IDDO data repository contains over 1.3 million IPD records from over 600 studies and 70
 115 countries, across 8 disease themes. Since February 2026, all researchers accessing data from
 116 the IDDO data repository have been provided with information on how to use and access the
 117 `iddoverse` package, thus increasing the number and diversity of users and organisations.

118 Research which has used the `iddoverse` includes malaria studies conducted by the Liverpool
 119 School of Tropical Medicine and the Mahidol-Oxford Tropical Medicine Research Unit in
 120 Thailand (unpublished at time of writing). Published works which have used the `iddoverse`
 121 package include research on factors associated with death from Ebola (Yeabah TO, 2025) and
 122 two visceral leishmaniasis meta-analyses (Kumar R, 2023; Munir A, 2023). Additionally, the
 123 package is being used internally by IDDO to create summaries to check, and subsequently
 124 improve, the quality of the data standardisation.

125 In a survey conducted by IDDO, several users of the IDDO data repository have reported
 126 confusion with the variety of timing variables and result columns. Users will benefit from the

127 iddoverse package as a solution to this complexity.

128 Acknowledgements

129 This research was supported by the Wellcome Trust [222410/Z/21/Z].

130 Special thanks to Dr Caitlin Naylor for their project management during the project.

131 AI Usage Disclosure

132 Artificial Intelligence (AI) models were not used in documentation writing, paper authoring or
133 dataset generation. ChatGPT 5.2 (OpenAI, 2025) was used to suggest unit tests using R code
134 for the development of testthat script files. This code was manually verified for quality and
135 correctness, tested and modified before adding to the package. AI models were not used in
136 the code creation for any of the exported functions in the package.

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